THE GYSIN MAPS OF COMPOSABLE LOCAL COMPLETE INTERSECTION MORPHISMS COMPOSE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FF5 (sources/stacks-chow/pinned-original-source.tex line 13575).

Allowed standard prerequisites (editorial).

- Composition of local complete intersection morphisms. More on Morphisms, Lemma labelled more-morphisms-lemma-composition-lci, cited at proof lines 13583-13585. Makes the statement meaningful.
- The composition formula for the Gysin maps of regular immersions. Same chapter, Lemma labelled lemma-composition-regular-immersion, cited at proof lines 13608-13609. Supplies the composition of the immersion parts.
- Composition of flat pullbacks. Same chapter, Lemma labelled lemma-compose-flat-pullback, cited at proof lines 13608-13609. Supplies the composition of the smooth parts.
- Existence of the Gysin map through an immersion into a smooth scheme. Same chapter, Definition labelled definition-lci-gysin, used at proof lines 13587-13590. Proves that f^! exists.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 069J: sources/stacks-chow/tags.txt, source lines 8043–8043. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13585 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FF0: sources/stacks-chow/pinned-original-source.tex, source lines 13412–13412. Same-chapter pointer into the author environment lemma

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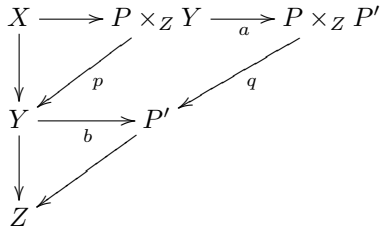
at source line 13412; invoked at proof line 13606 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02RD: sources/stacks-chow/pinned-original-source.tex, source lines 2287–2287. Same-chapter pointer into the author environment lemma at source line 2287; invoked at proof line 13607 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let $f : X \to Y$ and $g : Y \to Z$ be local complete intersection morphisms of schemes locally of finite type over S. Assume the Gysin map exists for $g \circ f$ and g. Then the Gysin map exists for f and $(g \circ f)^! = f^! \circ g^!$.*

Proof. Observe that $g \circ f$ is a local complete intersection morphism by More on Morphisms, Lemma 069J and hence the statement of the lemma makes sense. If $X \to P$ is an immersion of X into a scheme P smooth over Z then $X \to P \times_Z Y$ is an immersion of X into a scheme smooth over Y. This prove the first assertion of the lemma. Let $Y \to P'$ be an immersion of Y into a scheme P' smooth over Z. Consider the commutative diagram



Here the horizontal arrows are regular immersions, the south-west arrows are smooth, and the square is cartesian. Whence $a^! \circ q^* = p^* \circ b^!$ as bivariant classes commute with flat pullback. Combining this fact with Lemmas 0FF0 and 02RD the reader finds the statement of the lemma holds true. Small detail omitted. □